

Multiple Choice (10 marks)

1. Our eyes are the principal organs involved in seeing. Yet we are able to perform manual acts such as dressing or tying knots when our eyes are closed. Why?
 - (a) Gustatory receptors
 - (b) Photoreceptors
 - (c) Auditory receptors
 - (d) Proprioceptors
 - (e) Thermoreceptors
2. All of the following might be found in connective tissue EXCEPT
 - (a) thrombin
 - (b) glycosaminoglycans
 - (c) collagens
 - (d) fibroblasts
3. Vinegar is classified as an acid. What does this tell you about the concentration of the hydrogen ions and the hydroxide ions?
 - (a) Vinegar has a lower concentration of hydrogen ions than hydroxide ions
 - (b) Vinegar has an excess of hydrogen ions with respect to hydroxide ions
 - (c) Vinegar has a higher concentration of hydroxide ions than water
 - (d) Vinegar has no hydroxide ions
 - (e) Vinegar does not contain hydrogen ions
4. What is the function of the contractile vacuole?
 - (a) Protects the cell from foreign invaders
 - (b) Acts as a storage for genetic material
 - (c) Regulates the pH level within the cell
 - (d) Transports oxygen throughout the cell
 - (e) Removes nitrogenous wastes
5. If two modern organisms are distantly related in an evolutionary sense, then one should expect that
 - (a) their rate of mutation should be identical.
 - (b) they should have the exact same DNA structure.

- (c) they should have identical physical characteristics.
- (d) they should share the same number of chromosomes.
- (e) they live in very different habitats.

True / False (10 marks)

Answer True or False

- 6. True or False: Do general practice characteristics influence uptake of an information technology (IT) innovation in primary care?
- 7. True or False: Does the SCL 90-R obsessive-compulsive dimension identify cognitive impairments?
- 8. True or False: Nasal polyposis: is there an inheritance pattern?
- 9. True or False: Pitfalls in urinary stone identification using CT attenuation values: are we getting the same information on different scanner models?
- 10. True or False: Are reports of mechanical dysfunction in chronic oro-facial pain related to somatisation?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the potential hazards associated with molecular cloning, focusing on the implications for public health, biosecurity, and ethical concerns in genetic research.
- 12. Discuss the role of cyclic electron flow through photosystem I in eukaryotic photosynthetic cells and analyze its impact on the reduction of NADP⁺ to NADPH.

End of Paper